Lead 90th-percentile estimation from censored tap data - method

This note specifies the estimation method for the 90th percentile of lead and copper tap concentrations when some results are censored (detection limits). The data for each system, monitoring period, and contaminant are the reported sample values with their qualifiers. Only the roster in systems.csv defines which system-period cells are reported; results rows for other periods never count.

Unit conversion is per row: a value reported in ug/L is divided by 1000 to give mg/L. Each row converts by its own unit; the file carries mixed units and no column-wide assumption is valid.

A result is censored when its qualifier is U or begins with <, or when its converted concentration is zero or below. Other qualifiers (including J and B) are detects and are used at face value. A blank value field is unparseable and the row is skipped.

Only LEAD and COPPER rows are used; rows for any other contaminant are ignored.

Lognormal regression on order statistics (ROS)

Take the natural log of every detected concentration. Sort all observations (detects and censored limits) in joint ascending order by converted concentration. A censored limit ranks below a detect reported at the same numeric value, because the true concentration is at or below the limit.

Assign each rank i = 1..n the Cunnane plotting position p = (i - 0.4) / (n + 0.2) and the standard normal quantile z of p. Regress the log-detects on their normal quantiles (ordinary least squares): the fitted line gives the lognormal location and scale. The regression uses detects only; censored limits are never regressed, only imputed.

Impute each censored observation from the fitted line at its own rank quantile. The completed log-sample is the observed log-detects plus the imputed values. Its mean mu and sample standard deviation sigma (denominator n - 1) are the lognormal parameters.

The 90th percentile estimate is p90 = exp(mu + z_0.90 * sigma), where z_0.90 is the 0.90 quantile of the standard normal. Report p90 in mg/L rounded half up to four decimals.

Sampling uncertainty - one-sided upper confidence limit

The reported finding accounts for sampling uncertainty through a one-sided upper 95 percent confidence limit on the lognormal 90th percentile, by the delta method. Its standard error is se = sigma * sqrt(1/n + z_0.90^2 / (2*(n-1))). The upper limit is UCL95 = exp(mu + z_0.90 * sigma + z_0.95 * se), where z_0.95 is the 0.95 quantile of the standard normal. Report UCL95 in mg/L rounded half up to four decimals.

A cell exceeds the action level when its UCL95 is greater than the action level: 0.010 mg/L for lead, 1.3 mg/L for copper. The point estimate alone never decides the finding; a p90 below the action level with a UCL95 above it is an exceedance.

Degenerate and edge cells

An empty cell (no usable rows for the roster system, period, and contaminant) reports n = 0, n_censored = 0, p90 = 0.0000, UCL95 = 0.0000, and no exceedance.

A cell whose only usable rows are censored reports the minimum limit: p90 and UCL95 both equal the smallest censored concentration in mg/L.

A cell with exactly one detected value among any number of censored values reports the detected value for both p90 and UCL95. A cell with a single usable row (n = 1) reports that row’s concentration for both. A cell with zero fitted variance reports UCL95 = p90.

Rounded half-up worked check. Completed log values -5.80914, -5.65599, -5.52146, -5.29832, -5.11600 with mu = -5.48018 and sigma = 0.27506 give p90 = exp(-5.48018 + 1.28155 * 0.27506) = 0.0059 mg/L and UCL95 = 0.0087 mg/L at n = 10.